

Project Interim Report
On
File Compression Simulator
(Using Core Java & OOP Concepts)

RU-100-01-00012
Object Oriented Programming with Java

SESSION: 2025-26



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Abstract

The Automated Teller Machine (ATM) Simulation is a Java-based project developed using Object-Oriented Programming principles. The project simulates ATM functions such as PIN authentication, cash withdrawal, balance inquiry and secure account management. Encapsulation and exception handling are used to improve security and reliability.

Introduction

This project develops a console-based ATM Simulation using Java and Object-Oriented Programming concepts. The system authenticates users using a predefined PIN and provides withdrawal and balance inquiry facilities.

Scope

Suitable for educational purposes and small-scale banking simulations. It can be extended using databases, GUI, transaction history and multiple user accounts.

System Requirements

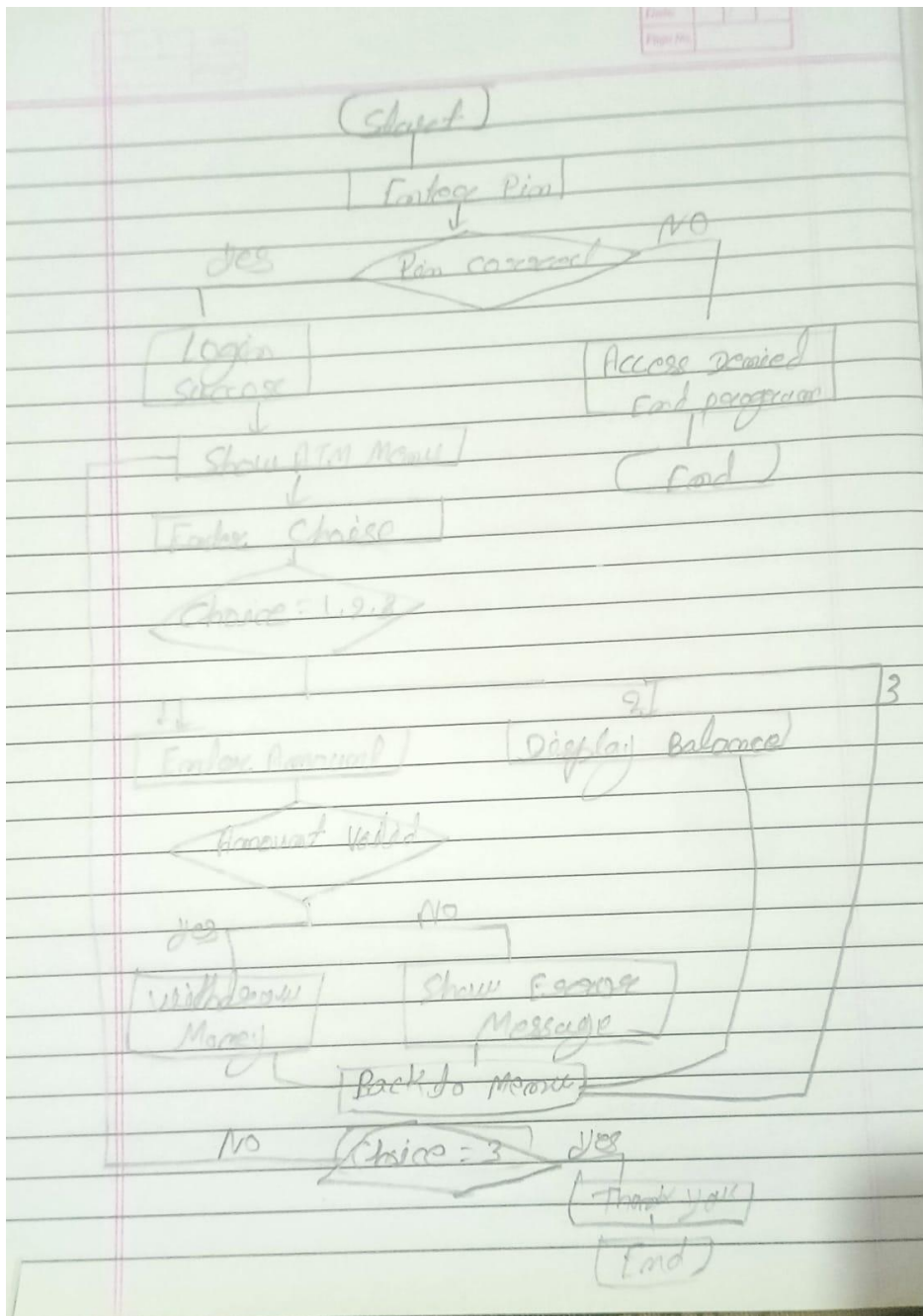
Hardware: Computer with 4GB RAM

Software: Java JDK, IDE (VS Code/Eclipse/Edit Plus), Windows

Methodology

1. Create Account and ATM objects.
2. Verify PIN.
3. Display menu.
4. Perform withdrawal/check balance.
5. Handle exceptions.
6. Exit when selected.

Flowchart



ClassName

Account(balance, withdraw(), getBalance())

ATM(PIN, authenticate(), showMenu())

ATMSimulation(main())

Implementation

Account class manages balance securely. ATM class handles authentication and menu display. ATMSimulation controls program execution and exception handling

```
> cd "c:\U
Enter PIN: 1234
Login Successful!

ATM MENU
1. Withdraw Money
2. Check Balance
3. Exit
Enter your choice: 2
Current Balance: ?2000.0

ATM MENU
1. Withdraw Money
2. Check Balance
3. Exit
Enter your choice: 1
Enter amount to withdraw: 200
Transaction successful.

ATM MENU
1. Withdraw Money
2. Check Balance
3. Exit
Enter your choice: 1
Enter amount to withdraw: [ ]
```

Gantt Chart

Week1-Planning, Week2-Design, Week3-Coding & Testing, Week4-Documentation

Conclusion

The project successfully demonstrates Java OOP concepts through a secure ATM simulation system.